

The equal-ratio monomial asymptotic in general dimension

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Abstract

Assembly of the general- d equal-ratio monomial milestone (Astra #14, units 165, 170–175). For $h, k: \text{Fin}(m+1) \rightarrow \mathbb{N}$ with $k_i > 0$ and common ratio $(h_i + 1)/(2k_i) = \lambda$, the monomial box integral

$$I(N) = \int_{(0,1]^{m+1}} \prod_i x_i^{h_i} e^{-\beta N \prod_i x_i^{2k_i}} dx$$

equals *exactly* $\prod_i \frac{1}{2k_i} \cdot \frac{1}{m!} \int_0^1 z^{\lambda-1} (-\log z)^m e^{-\beta N z} dz$ (`monomialBoxReal_eq`), hence

$$I(N) \sim \frac{\Gamma(\lambda)\beta^{-\lambda}}{m! \prod_i 2k_i} N^{-\lambda} (\log N)^m \quad (N \rightarrow \infty)$$

(`monomialBoxReal_isEquivalent'`), the logarithmic multiplicity $d = m + 1$ coming from the product density of unit 172 and the constant from the Gamma/log asymptotic of unit 173. Zero sorry/axiom.

1 The things formalised

```
def expKernel ( N : ) (z : ) : 0∞ := ofReal (exp (-( * N * z)))
theorem monomialBoxIntegral_exp_eq (m) (h k : Fin (m+1) → ) (hk) (1 N) (h1 : 0 < 1)
  (hratio : i, ((h i : ) + 1) / (2 * k i) = 1) (hN : 0 < * N) :
  monomialBoxIntegral (m + 1) h k (expKernel N)
    = ofReal (( i, 1 / (2 * k i)) * (1 / m.factorial) * gammaLogIntegral 1 m N)
def monomialBoxReal (d) (h k : Fin d → ) ( N : ) : :=
  x in unitBox d, ( i, x i ^ h i) * exp (-( * N * i, x i ^ (2 * k i)))
theorem ofReal_monomialBoxReal (d h k N) :
  ofReal (monomialBoxReal d h k N) = monomialBoxIntegral d h k (expKernel N)
theorem monomialBoxReal_eq ... : monomialBoxReal (m + 1) h k N
  = ( i, 1 / (2 * k i)) * (1 / m.factorial) * gammaLogIntegral 1 m N
theorem monomialBoxReal_tendsto ... : Tendsto
  (fun N => monomialBoxReal (m + 1) h k N / (N ^ (-1) * log N ^ m)) atTop
  ( (( i, 1 / (2 * k i)) * (1 / m.factorial) * (Gamma 1 * ^ (-1))))
theorem monomialBoxReal_isEquivalent ... : (fun N => monomialBoxReal (m + 1) h k N)
  ~[atTop] powLog (( i, 1 / (2 * k i)) * (1 / m.factorial) * (Gamma 1 * ^ (-1))) 1 m
theorem monomialBoxReal_isEquivalent' ... : (fun N => monomialBoxReal (m + 1) h k N)
  ~[atTop] fun N => Gamma 1 * ^ (-1) / (m.factorial * i, 2 * k i) * N ^ (-
1) * log N ^ m
```

2 Mathematics

Unit 174 reduces the monomial integral to the weighted box integral with weights $(h_i + 1)/(2k_i) - 1 = \lambda - 1$ and Jacobian $\prod_i 1/(2k_i)$; unit 172 identifies the constant-weight box integral with the

recursive product integral and evaluates it through the density $z^{\lambda-1}(-\log z)^m/m!$; unit 173 supplies the asymptotic of the resulting one-dimensional integral. The Bochner integral is bridged to the $[0, \infty]$ -valued one by `ofReal_integral_eq_lintegral_ofReal`: the integrand is continuous, hence integrable on the compact closed box $[0, 1]^d$ and a fortiori on $(0, 1]^d$, and nonnegative there.

3 Paper-facing meaning

This is the general-dimensional statement behind the equal-ratio case of the Taylor-tree multiplicity: the leading exponent is the common Mellin ratio λ and the logarithmic multiplicity is the number of coordinates realising it. Compared with the quadratic-chart multiplicity theorem of units 22–60 (`boxIntegral_isEquivalent`, kernel $e^{-\beta s^2 + \beta a s}$ with $s = \sqrt{n} \prod u_i^{k_i}$), the present statement covers an arbitrary monomial exponent $e^{-\beta N \prod x_i^{2k_i}}$ with the Gamma-function constant $\Gamma(\lambda)\beta^{-\lambda}$ in place of the weighted mass. Astra #14’s scope sentence now reads: general logarithmic multiplicity is complete both for the recursive quadratic chart and for arbitrary equal-ratio monomial charts.

4 Lean notes

- `Tendsto.div_const` needs the constant explicitly when the goal is not yet determined (placeholder `y` error).
- Membership in `Set.pi univ t` unfolds definitionally: `hx i (mem_univ i)` gives `x i t i`.
- Continuity of a finite product: `continuous_finsetProd _ fun i _ => ...` (`fun_prop` was not tried for the `Finset` product; the manual term is short).
- `ENNReal.ofReal_prod_of_nonneg` turns `ofReal ( f i)` into `ofReal (f i)`; the remaining goal is `rfl` once `expKernel` is unfolded.